

ASSIGNMENT NO. 5: Aggregate Functions and Data Analysis

Q1) Create a MySQL database named CollegeDB and a table named StudentResult.

The table should contain the following columns:

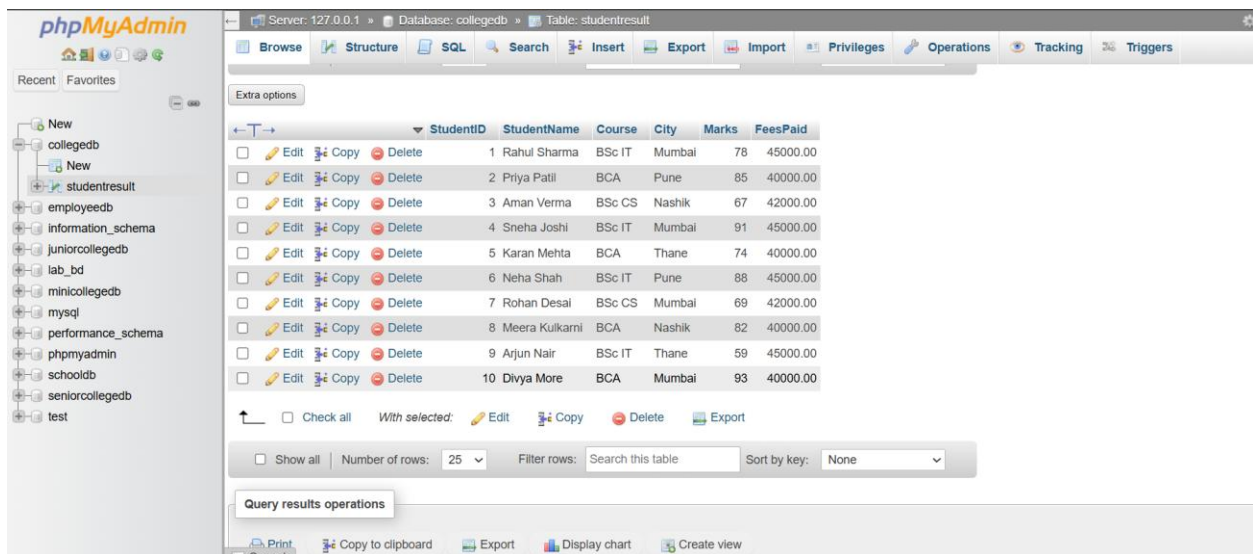
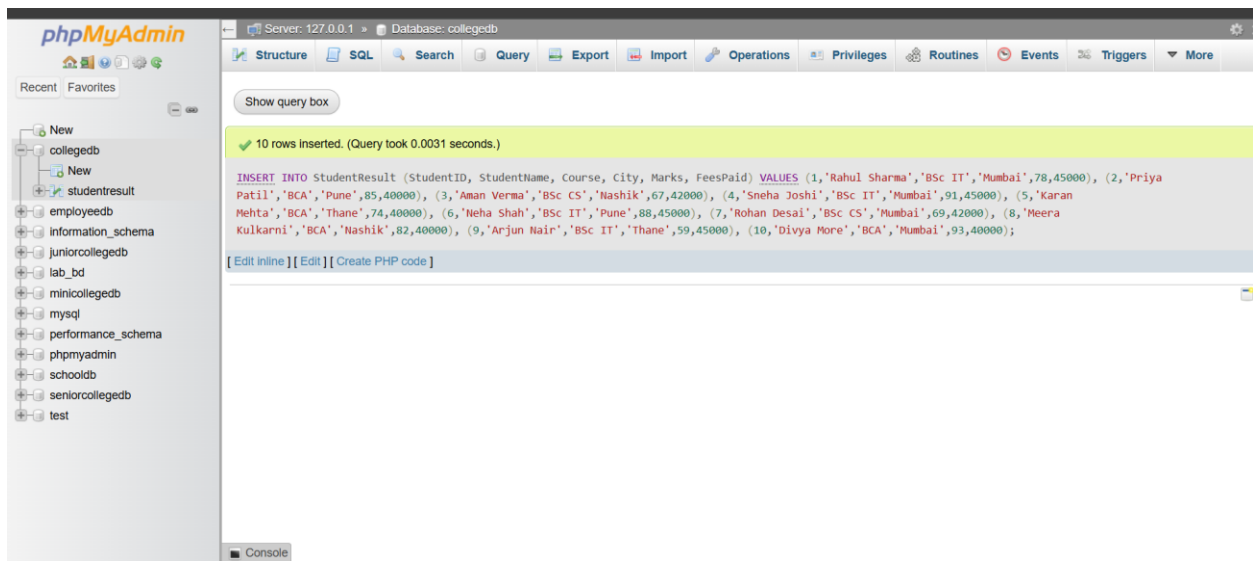
Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	
Course	VARCHAR(30)	
City	VARCHAR(30)	
Marks	INT	
FeesPaid	DECIMAL(10,2)	

Insert at least 10 records into the table.

```
CREATE DATABASE CollegeDB;  
USE CollegeDB;
```

```
CREATE TABLE StudentResult (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50),  
    Course VARCHAR(30),  
    City VARCHAR(30),  
    Marks INT,  
    FeesPaid DECIMAL(10,2)  
);
```

```
INSERT INTO StudentResult (StudentID, StudentName, Course, City, Marks,  
FeesPaid)  
VALUES  
(1, 'Rahul Sharma', 'BSc IT', 'Mumbai', 78, 45000.00),  
(2, 'Priya Patil', 'BCA', 'Pune', 85, 40000.00),  
(3, 'Aman Verma', 'BSc CS', 'Nashik', 67, 42000.00),  
(4, 'Sneha Joshi', 'BSc IT', 'Mumbai', 91, 45000.00),  
(5, 'Karan Mehta', 'BCA', 'Thane', 74, 40000.00),  
(6, 'Neha Shah', 'BSc IT', 'Pune', 88, 45000.00),  
(7, 'Rohan Desai', 'BSc CS', 'Mumbai', 69, 42000.00),  
(8, 'Meera Kulkarni', 'BCA', 'Nashik', 82, 40000.00),  
(9, 'Arjun Nair', 'BSc IT', 'Thane', 59, 45000.00),  
(10, 'Divya More', 'BCA', 'Mumbai', 93, 40000.00);
```



Q2) Write MySQL queries to find the following from the StudentResult table:

Requirement
Total number of students
Total fees paid
Average marks
Lowest marks
Highest marks

```
SELECT
    COUNT(*) AS TotalStudents,
    SUM(FeesPaid) AS TotalFeesPaid,
    AVG(Marks) AS AverageMarks,
    MIN(Marks) AS LowestMarks,
    MAX(Marks) AS HighestMarks
FROM StudentResult;
```

ANS:

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Server: 127.0.0.1 » Database: collegedb » Table: StudentResult

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Your SQL query has been executed successfully.

```
SELECT COUNT(*) AS TotalStudents FROM StudentResult;
```

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Extra options

TotalStudents
10

Query results operations

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Server: 127.0.0.1 » Database: collegedb » Table: StudentResult

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⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

✔ Showing rows 0 - 0 (1 total, Query took 0.0015 seconds.)

```
SELECT AVG(Marks) AS AverageMarks FROM StudentResult;
```

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Extra options

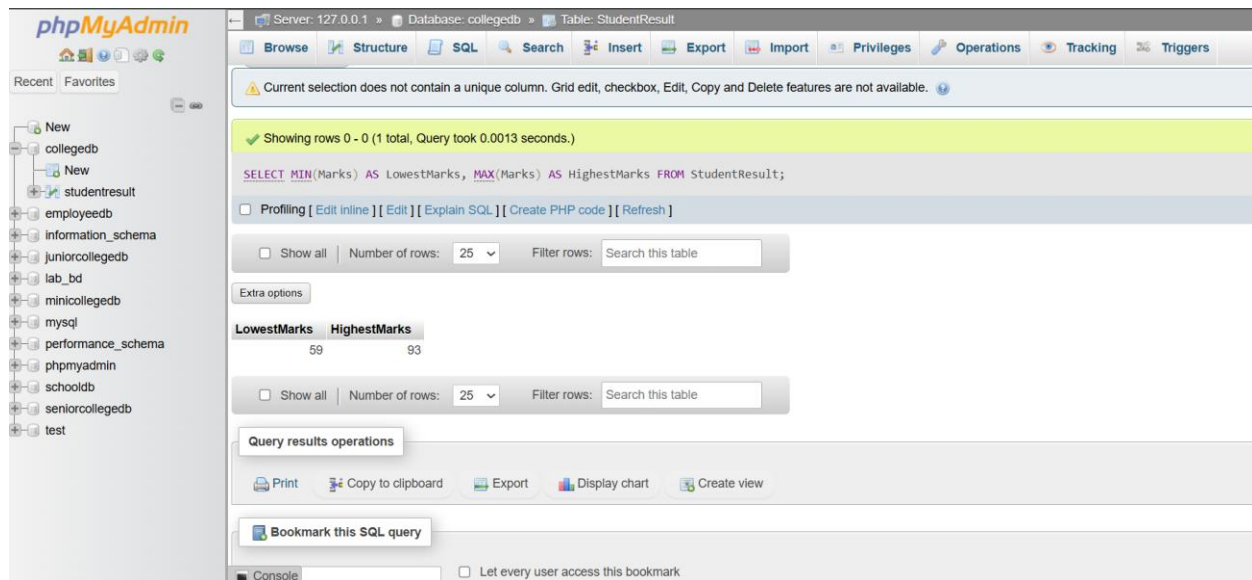
AverageMarks
78.6000

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Query results operations

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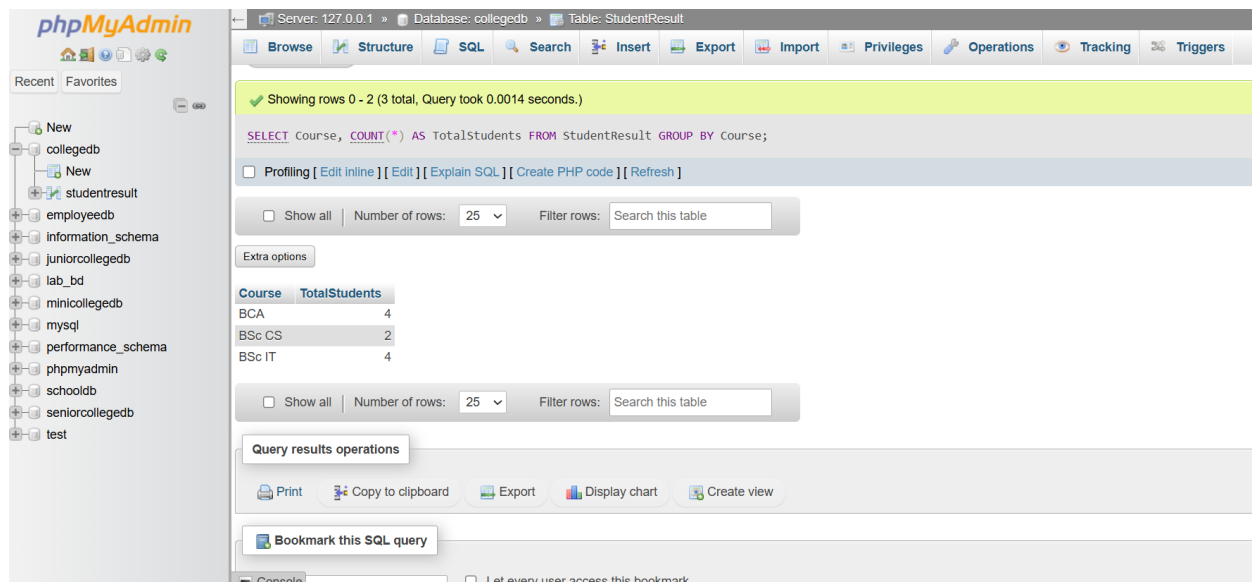


Q3) Write a MySQL query to count the number of students in each course.

The output should contain:

Column
Course
TotalStudents

```
SELECT Course, COUNT(*) AS TotalStudents
FROM StudentResult
GROUP BY Course;
```

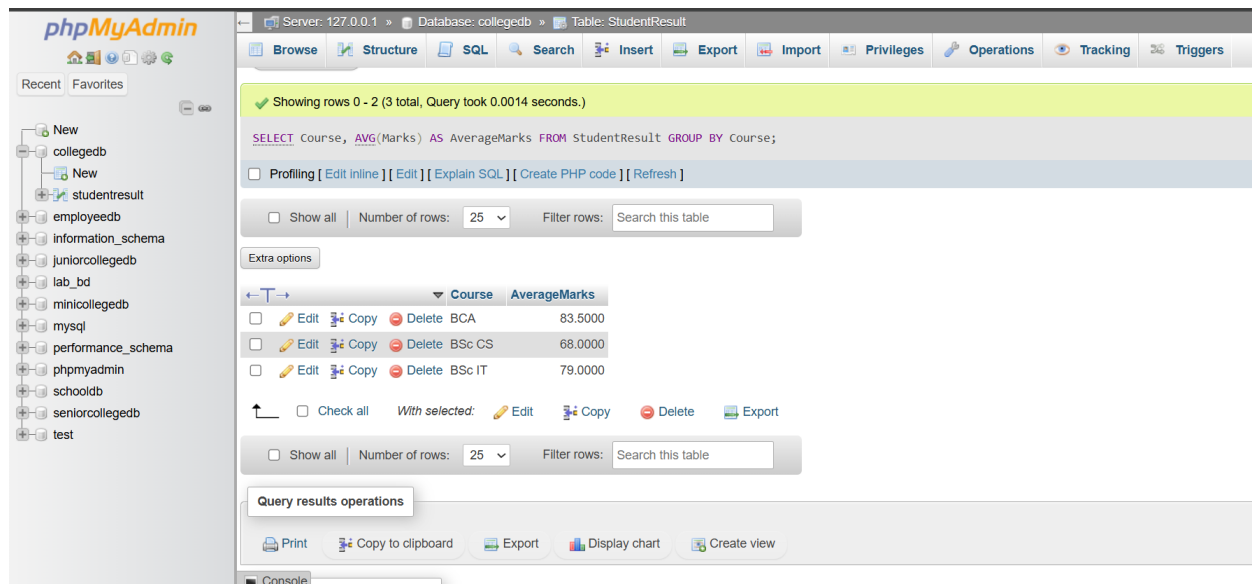


Q4) Write a MySQL query to find the average marks of students course-wise.

The output should contain:

Column
Course
AverageMarks

```
SELECT Course, AVG(Marks) AS AverageMarks
FROM StudentResult
GROUP BY Course;
```



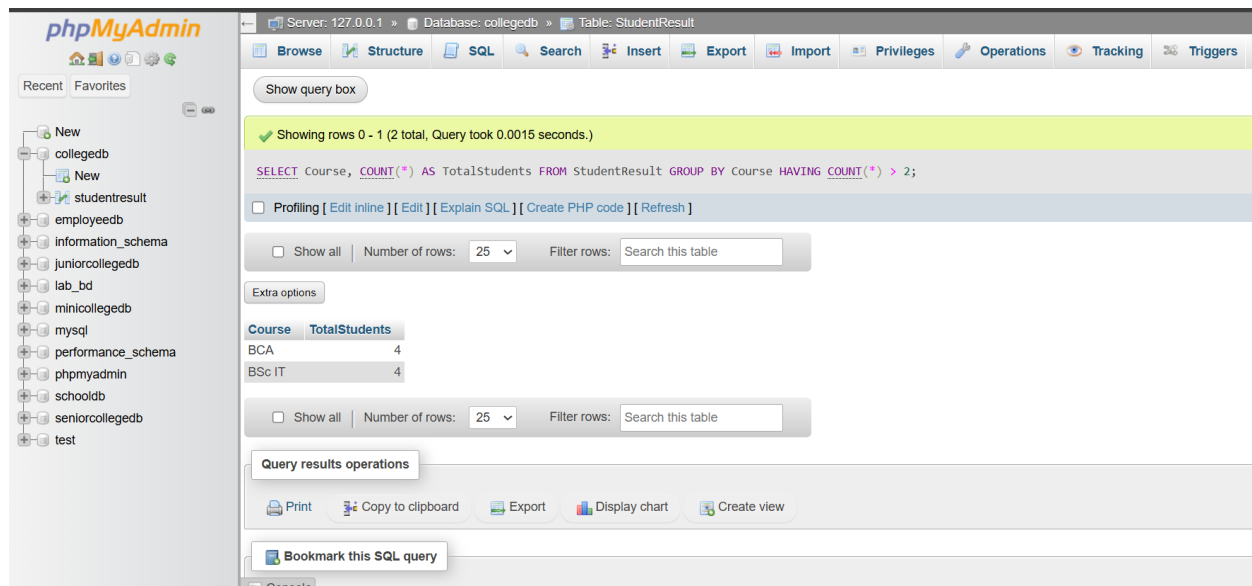
Course	AverageMarks
BCA	83.5000
BSc CS	68.0000
BSc IT	79.0000

Q5) Write a MySQL query to display only those courses where the number of students is greater than 2.

Use:

Clause
GROUP BY
HAVING

```
SELECT Course, COUNT(*) AS TotalStudents
FROM StudentResult
GROUP BY Course
HAVING COUNT(*) > 2;
```



Q6) Write a MySQL query to analyze course performance.

The output should display:

Column
Course
TotalStudents
AverageMarks
LowestMarks
HighestMarks

Sort the result by AverageMarks in descending order.

```

SELECT
    Course,
    COUNT(*) AS TotalStudents,
    AVG(Marks) AS AverageMarks,
    MIN(Marks) AS LowestMarks,
    MAX(Marks) AS HighestMarks
FROM StudentResult
GROUP BY Course
ORDER BY AverageMarks DESC;

```

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test

Server: 127.0.0.1 » Database: collegedb » Table: StudentResult

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Showing rows 0 - 2 (3 total, Query took 0.0014 seconds.)

SELECT Course, COUNT(*) AS TotalStudents, SUM(FeesPaid) AS TotalFees, AVG(Marks) AS AverageMarks, MIN(Marks) AS LowestMarks, MAX(Marks) AS HighestMarks FROM StudentResult GROUP BY Course ORDER BY AverageMarks DESC;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all

Number of rows: 25

Filter rows: Search this table

Extra options

Course	TotalStudents	TotalFees	AverageMarks	LowestMarks	HighestMarks
BCA	4	160000.00	83.5000	74	93
BSc IT	4	180000.00	79.0000	59	91
BSc CS	2	84000.00	68.0000	67	69

Show all

Number of rows: 25

Filter rows: Search this table

Query results operations

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